Arista Access Node MAR-ACC-04 QoS Congestion Event SOP

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1419

Description

The Marathahalli Core site's Access Node MAR-ACC-04, a critical juncture in the network topology, is experiencing a QoS congestion event due to microbursts and interface tail drops, resulting in ALARM_1419. This phenomenon can be attributed to the inherent characteristics of Arista's EOS operating system and the specific role of the Access Node in handling a high volume of traffic. The QoS configuration on the Access Node is set to prioritize critical traffic flows, but the occurrence of microbursts, which are transient periods of extremely high traffic intensity, can overwhelm the QoS queues, leading to tail drops and subsequent congestion. Furthermore, the Access Node's role in aggregating traffic from multiple ports can exacerbate the issue, creating a bottleneck in the network. The combination of these factors necessitates a comprehensive troubleshooting approach to identify and resolve the root cause of the QoS congestion event.

Pre-requisite Checks

- Verify the site location and node ID to ensure the correct node is being addressed.
- Confirm the severity of the alarm and the impact on the network.
- Check the redundant link status to ensure it is operational and capable of handling the traffic load.
- Dispatch remote hands to Marathahalli Core to facilitate access to the node.

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops due to QoS queue starvation	High
Inadequate QoS configuration on the Access Node	Medium
Hardware or software issues with the Access Node or its components	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System Resource Usage, Description: This command displays the system resource usage, including CPU, memory, and disk usage. It helps identify if there are any hardware or software issues contributing to the QoS congestion event.

Command: show qos queue statistics, Expected Output: QoS Queue Statistics, Description: This command displays the QoS queue statistics, including the number of packets dropped, tail drops, and queue occupancy. It helps identify if there are any QoS queue starvation issues contributing to the congestion event.
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Command: show interface statistics, Expected Output: Interface Statistics, Description: This command displays the interface statistics, including the number of packets sent, received, and dropped. It helps identify if there are any interface-related issues contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify QoS configuration, Command: show qos configuration, Description: Verify the QoS configuration on the Access Node to ensure it is correctly configured to prioritize critical traffic flows.

Step 2: Step: Adjust QoS configuration, Command: configure qos queue, Description: Adjust the QoS configuration to increase the queue size and adjust the priority of critical traffic flows to mitigate the effects of microbursts and interface tail drops.

Step 3: Step: Verify interface settings, Command: show interface settings, Description: Verify the interface settings on the Access Node to ensure they are correctly configured to handle the traffic load.

Post-Check Verification

```
Command: show qos queue statistics, Expected Output: QoS Queue Statistics, Description: Verify that the QoS queue statistics indicate a reduction in packets dropped and tail drops.

Command: show interface statistics, Expected Output: Interface Statistics, Description: Verify that the interface statistics indicate a reduction in packets dropped.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: configure qos configuration default, Description: Restore the original QoS configuration to ensure the Access Node is correctly configured to prioritize critical traffic flows.

- Step: Verify QoS configuration, Command: show qos configuration, Description: Verify that the QoS configuration is correctly restored.